Chemical Bonding

Chemistry revision guide

Study snapshot

Review how atoms form bonds and how structure influences molecular properties.

Key ideas

- Ionic bonding involves attraction between oppositely charged ions.
- Covalent bonding involves shared pairs of electrons.
- Lewis structures show valence electrons and bonding pairs.
- Electron-pair repulsion helps predict simple molecular shapes.
- Polarity depends on bond polarity and the overall molecular shape.

Try the questions on page 2 before checking your work.

Chemical Bonding

Quick practice

1. What type of bond forms between Na+ and Cl-?

2. How many shared electron pairs are in a typical double bond?

3. What is the approximate shape of a molecule with two bonding pairs and no lone pairs around the central atom?

4. Can a molecule with polar bonds be non-polar overall?

Answer notes

1) Ionic attraction
2) Two shared pairs
3) Linear
4) Yes, if the bond dipoles cancel by symmetry

Review the method, then retry any question you missed.